SUPPORT VECTOR MACHINE – REGRESSION

[2]:
```
import pandas as pd
import pickle
import matplotlib.pyplot as plt
```

[10]:
```
dataset = pd.read_csv('Startups_dataset.csv')
dataset
```

[10]:

	Unnamed: 0	R&D Spend	Administration	Marketing Spend	State	Profit
0	0	165349.30	136897.90	471784.20	New York	192261.93
1	1	162597.80	151377.69	443898.63	California	191792.16
2	2	153441.61	101145.65	407934.64	Florida	191050.49
3	3	144372.51	118671.95	383199.72	New York	182902.09
4	4	142107.44	91391.87	366168.52	Florida	166188.04
5	5	131877.00	99814.81	362861.46	New York	156991.22
6	6	134615.56	147198.97	127716.92	California	156122.61
7	7	130298.23	145530.16	323876.78	Florida	155752.70
8	8	120542.62	148719.05	311613.39	New York	152211.87

[11]:
```
dataset.columns
```

[11]:
```
Index(['Unnamed: 0', 'R&D Spend', 'Administration', 'Marketing Spend', 'State',
       'Profit'],
      dtype='str')
```

[12]:
```
# Apply one-hot encoding
dataset = pd.get_dummies(dataset, columns=['State'], drop_first=True)
dataset
```

[12]:

	Unnamed: 0	R&D Spend	Administration	Marketing Spend	Profit	State_Florida	State_New York
0	0	165349.30	136897.90	471784.20	192261.93	False	True
1	1	162597.80	151377.69	443898.63	191792.16	False	False
2	2	153441.61	101145.65	407934.64	191050.49	True	False
3	3	144372.51	118671.95	383199.72	182902.09	False	True
4	4	142107.44	91391.87	366168.52	166188.04	True	False
5	5	131877.00	99814.81	362861.46	156991.22	False	True
6	6	134615.56	147198.97	127716.92	156122.61	False	False
7	7	130298.23	145530.16	323876.78	155752.70	True	False
8	8	120542.62	148719.05	311613.39	152211.87	False	True

[13]:
```
dataset.columns
```

[13]:
```
Index(['Unnamed: 0', 'R&D Spend', 'Administration', 'Marketing Spend',
       'Profit', 'State_Florida', 'State_New York'],
      dtype='str')
```

[17]:
```
x = dataset[["R&D Spend", "Administration", "Marketing Spend", "State_Florida", "State_New York"]]
y = dataset[["Profit"]]
```

[19]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.20,random_state=24)
```

[20]:
```
x_train
```

[20]:

	R&D Spend	Administration	Marketing Spend	State_Florida	State_New York
20	76253.96	113867.40	298664.57	False	False
33	55494.05	103057.59	214634.91	True	False
27	72107.70	127864.65	353183.91	False	True
30	61994.58	115641.38	91131.34	True	False
10	101913.18	110594.21	229161.05	True	False
49	0.10	116983.90	45173.16	False	False
42	23641.03	96189.73	148001.21	False	False
21	78389.57	153773.53	299737.39	False	True
14	119943.34	156547.52	256513.02	True	False

[22]:
```
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
x_train=sc.fit_transform(x_train)
x_test=sc.transform(x_test)
```

[126]:
```
from sklearn.svm import SVR
reg = SVR(kernel = "linear", gamma = 'auto', C = 1800)
reg.fit(x_train,y_train)
```

```
C:\Users\Admin\anaconda3\Lib\site-packages\sklearn\utils\validation.py:1365: DataConversionWarn
expected. Please change the shape of y to (n_samples, ), for example using ravel().
  y = column_or_1d(y, warn=True)
```

[126]:

SVR

▸ Parameters

▸ Fitted attributes

[127]:
```
y_pred = reg.predict(x_test)
y_pred
```

[127]:
```
array([150014.63811637, 157274.68801453,  92786.54763134, 107338.44261103,
       106817.13921787, 170614.81323636, 117382.7042733 ,  81690.28458401,
       102907.84495074,  87990.39918913])
```

[128]:
```
from sklearn.metrics import r2_score
score = r2_score(y_test,y_pred)
score
```

[128]:
```
0.9272830923533087
```

KERNAL	GAMMA	C	R2_SCORE
linear	scale	10	-0.150508906
linear	scale	100	0.073532952
linear	scale	1000	0.813909249
linear	scale	1800	0.927283092
linear	auto	10	-0.150508906
linear	auto	100	0.073532952
linear	auto	1000	0.813909249
linear	auto	1800	0.927283092
rbf	scale	10	-0.179848656
rbf	scale	100	-0.164299084
rbf	scale	1000	-0.047132937
rbf	scale	1800	0.03183102
rbf	auto	10	-0.179848656
rbf	auto	100	-0.164299084
rbf	auto	1000	-0.047132937
rbf	auto	1800	0.03183102
sigmoid	scale	10	-0.176213043
sigmoid	scale	100	-0.133434947
sigmoid	scale	1000	0.171474481
sigmoid	scale	1800	0.322328488
sigmoid	auto	10	-0.176213043
sigmoid	auto	100	-0.133434947
sigmoid	auto	1000	0.171474481
sigmoid	auto	1800	0.322328488
poly	scale	10	-0.175841455
poly	scale	100	-0.126111297
poly	scale	1000	0.276251754
poly	scale	1800	0.484751549
poly	auto	10	-0.175841455
poly	auto	100	-0.126111297
poly	auto	1000	0.276251754
poly	auto	1800	0.484751549